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Roll no : 12

Section : B4

Batch : B1

Aim : Write a python program to demonstrate the working K-Nearest neighbors (KNN) classifier algorithm. Consider Diabetes dataset to predict whether a patient is diabetic or not.

```
from google.colab import files
uploaded=files.upload()
```

diabetes.csv
diabetes.csv(text/csv) - 23875 bytes, last modified: 3/18/2026 - 100% done
Saving diabetes.csv to diabetes.csv

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('diabetes.csv')
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
```

```
df.isnull().sum()
```

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

```
x = df.drop('Outcome', axis=1)
y = df['Outcome']
```

```
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
```

```
x_train, x_test, y_train, y_test = train_test_split(
x_scaled, y, test_size=0.2, random_state=42)
```

```
k = 5
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(x_train, y_train)
```

▼ KNeighborsClassifier ⓘ ?

KNeighborsClassifier()

```
from sklearn.metrics import confusion_matrix
print("Confusion Matrix:")
y_pred = knn.predict(x_test)
print(confusion_matrix(y_test, y_pred))
```

Confusion Matrix:

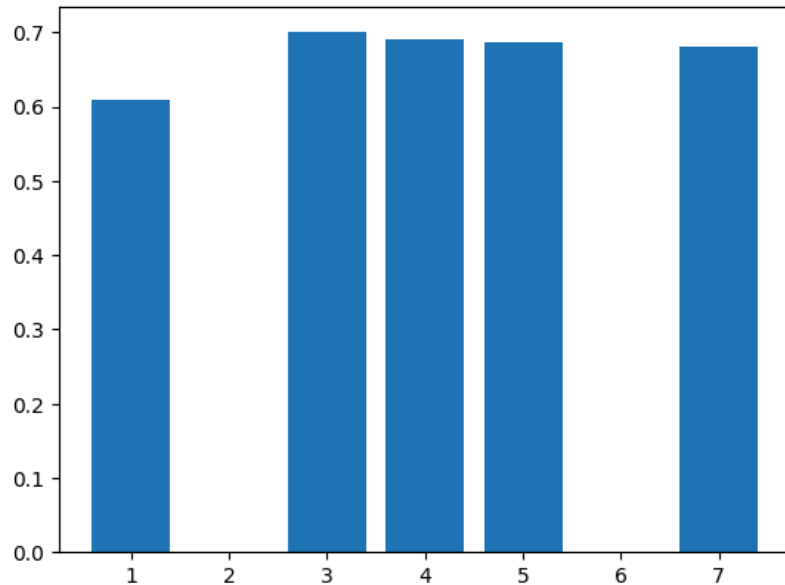
```
[[79 20]
 [28 27]]
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
```

```
print(f"Accuracy: {accuracy}")
print(f"Precision: {precision}")
print(f"Recall: {recall}")
print(f"F1 Score: {f1}")
```

```
Accuracy: 0.6883116883116883
Precision: 0.574468085106383
Recall: 0.4909090909090909
F1 Score: 0.5294117647058824
```

```
import matplotlib.pyplot as plt
k_values=[1, 3, 4, 5, 7]
accuracies=[0.61, 0.70, 0.69, 0.687, 0.68]
plt.bar(k_values, accuracies)
plt.show()
```



```
k = 5
knn = KNeighborsClassifier(n_neighbors=k)

# Train model
knn.fit(x_train, y_train)
```

```
# Predict
y_pred = knn.predict(x_test)
```

```
from google.colab import files
uploaded=files.upload()
```

No file chosen

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('tested (2).csv')
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
```

```
df.isnull().sum()
```

```
df.describe()
```

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# Separate Features and target
x = df.drop('Survived', axis=1)
y = df['Survived']

# Drop non-numeric columns that are not suitable for encoding or directly useful
x = x.drop(['Name', 'Ticket', 'PassengerId', 'Cabin'], axis=1, errors='ignore')

# One-hot encode categorical features
x = pd.get_dummies(x, columns=['Sex', 'Embarked'], drop_first=True)

# Feature scaling
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)

# Split dataset
x_train, x_test, y_train, y_test = train_test_split(
    x_scaled, y, test_size=0.2, random_state=42)
```

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